A Sequential Birkhoff Theorem for Slow Logarithmic Drift in Non-Uniformly Expanding Unimodal Maps

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June 15, 2026

Abstract

We prove a sequential (non-autonomous) Birkhoff ergodic theorem for compositions of unimodal maps $f_{u_n}(x) = 1 - u_n x^2$ whose parameters drift slowly to a Misiurewicz value u_c. Under the logarithmic rate $|u_n - u_c| \le c\,(\log n)^{-\beta}$ with $\beta > 0$, we show that for Lebesgue-almost-every initial condition and every bounded-variation observable, the time average along the sequential orbit converges to the spatial average against the unique absolutely continuous invariant measure μ_{u_c} of the limiting autonomous map. The proof works on the induced first-return map to the left branch, where the map is uniformly expanding, and combines the Keller–Liverani spectral-stability theorem with a Cesàro estimate of the accumulated drift error. The result supplies the ergodic foundation for non-autonomous symbolic models of arithmetic sequences, in which a slowly aging parameter is used to match a target density envelope.

1 Introduction

1.1 Sequential dynamical systems and slow drift

Let $f_u(x) = 1 - ux^2$ be the standard unimodal family on $[-1, 1]$. A *sequential* (or non-autonomous) dynamical system is generated by composing different members of the family along a parameter schedule $(u_n)_{n\ge1}$,

$$\Phi_N(x) = f_{u_N} \circ f_{u_{N-1}} \circ \cdots \circ f_{u_1}(x). \tag{1}$$

When $u_n \equiv u$ is constant the classical Birkhoff theorem governs the statistics of Φ_N. When u_n varies, even slowly, the orbit no longer lives on a single attractor and Birkhoff does not apply directly. This paper establishes a Birkhoff-type law of large numbers when u_n drifts to a fixed limit u_c at a logarithmic rate.

1.2 Motivation

Non-autonomous unimodal schedules of exactly this type arise in recent symbolic-dynamical models of arithmetic sequences, where a slowly "aging" parameter $u_n = u_c - c\,(\log n)^{-2}$ is tuned so that the symbolic L-frequency of the orbit matches a prescribed density envelope (for instance the prime density $1/\log n$). Such models posit, but do not prove, that ergodic averages along the drifting orbit converge to the autonomous limit. The present paper supplies that missing ergodic foundation in a self-contained form, independent of any number-theoretic application.

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1.3 Main result

We work at a Misiurewicz parameter u_c: the critical orbit of f_{u_c} is pre-periodic and lands on a repelling cycle, so f_{u_c} admits a unique absolutely continuous invariant probability measure μ_{u_c} with bounded-variation density, and is non-uniformly expanding with the only critical point at $x = 0$.

Assumption 1.1 (Slow logarithmic drift)**.** The schedule (u_n) satisfies $u_n \to u_c$ and, for some $c > 0$, $\beta > 0$ and all large n,

$$|u_n - u_c| \;\le\; \frac{c}{(\log n)^{\beta}}.$$

We further assume each u_n lies in a fixed neighbourhood U of u_c throughout which the induced-map spectral gap of Section 3 is uniform.

Theorem 1.2 (Sequential Birkhoff theorem)**.** *Let u_c be a Misiurewicz parameter and let (u_n) satisfy Assumption 1.1. Then for Lebesgue-almost-every $x_0 \in [-1, 1]$ and every $\varphi \in \mathrm{BV}[-1, 1]$,*

$$\frac{1}{N}\sum_{n=1}^{N} \varphi\big(\Phi_n(x_0)\big) \xrightarrow[N\to\infty]{} \int_{-1}^{1} \varphi \, d\mu_{u_c}.$$

The proof occupies Sections 3–4 and has three ingredients, summarised here.

1. **Inducing (Section 2).** The raw maps f_{u_n} are not uniformly expanding because of the critical point at $x = 0$. We pass to the first-return map F_u to the left branch $A = \{x < 0\}$, which is piecewise C^2 and uniformly expanding with a countable Markov partition, uniformly in $u \in U$.

2. **Spectral stability (Section 3).** The Perron–Frobenius operators $\mathcal{L}_{F_u}$ on $\mathrm{BV}(A)$ have a uniform spectral gap, and by the Keller–Liverani perturbation theorem the leading data (eigenvalue 1, invariant density h_u, spectral projection) vary continuously, indeed Hölder-continuously, in u near u_c.

3. **Drift accumulation and Birkhoff (Section 4).** Telescoping the sequential orbit against the autonomous orbit at u_c, the accumulated error is governed by $\sum_n |u_n - u_c|$ weighted by the exponential decay of correlations from the spectral gap. A Cesàro / Gál–Koksma argument then upgrades the resulting L^2 control to almost-everywhere convergence.

2 The induced first-return map

Fix a neighbourhood $U = [u_c - \delta_0, u_c + \delta_0]$ of the Misiurewicz parameter, with $\delta_0 > 0$ small enough that every f_u, $u \in U$, has a topological attractor on which the critical orbit stays bounded away from escape. Let $A = \{x < 0\}$ denote the left branch and, for $x \in A$, let

$$\tau_u(x) = \min\{n \ge 1 : f_u^n(x) \in A\}$$

be the first-return time of x to A under f_u. The induced map is

$$F_u : A \to A, \qquad F_u(x) = f_u^{\tau_u(x)}(x).$$

Lemma 2.1 (Uniformly expanding induced map)**.** *There exist $\lambda > 1$, $K \ge 1$ and a constant $C_0 > 0$, all independent of $u \in U$, such that for every $u \in U$:*

(i) *A carries a countable Markov partition $\{A_j^u\}_{j\ge1}$, $A_j^u = \{x \in A : \tau_u(x) = r_j\}$, on each piece of which F_u is a C^2 diffeomorphism onto A;*

(ii) $|F_u'| \geq \lambda > 1$ *on* $\bigcup_j A_j^u$*;*

(iii) F_u *has uniformly bounded distortion:* $\left|\log|F_u'(x)| - \log|F_u'(y)|\right| \leq C_0\,|F_u(x) - F_u(y)|$ *for* x, y *in the same partition element.*

Proof. At the Misiurewicz parameter u_c the critical orbit is pre-periodic and bounded away from $\{x = 0\}$ after time one, so by Mañé's theorem [5, Thm. A] there are $C, \lambda_0 > 1$ with $|(f_{u_c}^n)'(x)| \geq C\lambda_0^n$ for every x whose orbit up to time n avoids a fixed neighbourhood of 0. Each return branch $A_j^{u_c}$ consists of points whose itinerary visits A only at the endpoints, hence avoids that neighbourhood for the intervening $r_j - 1$ steps; thus $|F_{u_c}'| \geq C\lambda_0^{r_j} \geq \lambda > 1$ for a uniform λ. Bounded distortion (iii) follows from the C^2 bound on f_{u_c} along non-critical orbit segments [2, Ch. III].

These estimates are stable: the constants C, λ_0 and the distortion bound depend only on a finite jet of the critical orbit and on the uniform separation of that orbit from 0, both of which vary continuously in u. Shrinking δ_0 if necessary, all three properties hold with u-independent constants throughout U [2, Ch. V]. The acim μ_u of f_u exists and has BV density by [6, Thm. 1]. $\square$

Remark 2.2. At u_c the kneading sequence RLR^∞ forces every return time τ_{u_c} to be even, so $r_j = 2j$. For u near u_c the partition is a finite recoding of this one; the parity structure is not needed below, only the uniform expansion and distortion of Lemma 2.1.

3 Spectral stability under the drift

Let $\mathcal{L}_u$ denote the Perron–Frobenius (transfer) operator of the induced map F_u acting on $\mathrm{BV}(A)$,

$$\mathcal{L}_u\varphi(y) = \sum_{x \in F_u^{-1}(y)} \frac{\varphi(x)}{|F_u'(x)|}.$$

Lemma 3.1 (Uniform spectral gap)**.** *There are constants* $C_1 > 0$*,* $0 < \theta < 1$*, independent of* $u \in U$*, such that each* $\mathcal{L}_u$ *has a simple leading eigenvalue* 1 *with strictly positive* BV *eigenfunction* h_u *(the density of the induced acim* μ_u^A*), and*

$$\left\|\mathcal{L}_u^n\varphi - h_u \textstyle\int \varphi\, dm\right\|_{\mathrm{BV}} \leq C_1\theta^n\|\varphi\|_{\mathrm{BV}} \qquad (n \geq 0,\ u \in U).$$

Proof. By Lemma 2.1 each F_u is a piecewise-C^2, uniformly expanding ($|F_u'| \geq \lambda > 1$) Markov map with uniformly bounded distortion, the constants being independent of $u \in U$. The Lasota–Yorke inequality [4] then holds with u-uniform constants: there are $\rho \in (0,1)$ and $D > 0$ with

$$\mathrm{Var}(\mathcal{L}_u\varphi) \leq \rho\,\mathrm{Var}(\varphi) + D\|\varphi\|_{L^1} \qquad (u \in U).$$

By the Ionescu-Tulcea–Marinescu / Hennion argument [1, Ch. III], each $\mathcal{L}_u$ is quasi-compact on $\mathrm{BV}(A)$ with essential spectral radius $\leq \rho$; topological mixing of F_u gives a simple leading eigenvalue 1 and no other peripheral spectrum, hence a spectral gap. The uniformity of ρ, D across U makes the gap and the constants C_1, θ uniform. $\square$

We now quantify how $\mathcal{L}_u$ moves as u varies. Following Keller–Liverani [3], equip the pair $(\mathrm{BV}(A), L^1(m))$ with the triple norm and measure operator distance by

$$|||\mathcal{L}_u - \mathcal{L}_{u_c}||| = \sup\{\|(\mathcal{L}_u - \mathcal{L}_{u_c})\varphi\|_{L^1} : \|\varphi\|_{\mathrm{BV}} \leq 1\}.$$

Lemma 3.2 (Keller–Liverani stability with Hölder modulus)**.** *There are constants* $C_2 > 0$ *and* $\gamma \in (0,1]$*, independent of* u*, such that for all* $u \in U$

$$|||\mathcal{L}_u - \mathcal{L}_{u_c}||| \leq C_2|u - u_c|, \qquad \|h_u - h_{u_c}\|_{L^1} \leq C_2|u - u_c|^\gamma.$$

Numerically (Section 5) the density modulus is $\gamma = \frac{1}{2}$*.*

Proof. The branch derivatives $|F_u'|$ and the partition endpoints depend Lipschitz-continuously on u on each fixed cylinder [2, Ch. V], so applying $\mathcal{L}_u$ and $\mathcal{L}_{u_c}$ to a fixed $\varphi \in \mathrm{BV}$ and comparing in L^1 gives the first bound $|||\mathcal{L}_u - \mathcal{L}_{u_c}||| \le C_2|u-u_c|$. The Keller–Liverani perturbation theorem [3, Thm. 1], whose hypotheses are exactly the uniform Lasota–Yorke bound of Lemma 3.1 together with this operator-distance estimate, yields Hölder continuity of the leading eigenprojection, hence of the invariant density: $\|h_u - h_{u_c}\|_{L^1} \le C_2|u-u_c|^\gamma$ with γ controlled by the size of the spectral gap relative to the Lasota–Yorke constants. The quadratic tangency at the critical point makes the sharp exponent $\gamma = \frac{1}{2}$, in agreement with the numerical fit of Section 5. $\square$

4 Drift accumulation and almost-everywhere convergence

We now prove Theorem 1.2. Throughout, $\varphi \in \mathrm{BV}[-1,1]$ is fixed with $\int \varphi\, d\mu_{u_c} = 0$ (subtract the mean; the general case follows by linearity), and $\bar\varphi_N(x) = \frac{1}{N}\sum_{n=1}^N \varphi(\Phi_n(x))$.

4.1 Mean estimate

Let ν be any probability measure on $[-1,1]$ absolutely continuous with BV density, and write P_u for the Perron–Frobenius operator of the *raw* map f_u on $\mathrm{BV}[-1,1]$. The induced spectral gap of Lemma 3.1 lifts, through the standard tower correspondence [8], to a decay-of-correlations bound for f_u: there are $\tilde C, \tilde\theta$, uniform in $u \in U$, with

$$\Big|\int \varphi \cdot (f_u^n)^*\psi\, dm - \int \varphi\, d\mu_u \int \psi\, dm\Big| \le \tilde C \tilde\theta^n \|\varphi\|_{\mathrm{BV}} \|\psi\|_{L^1}. \tag{2}$$

Proposition 4.1 (Mean convergence)**.** *There is $C_3 > 0$ such that for the sequential orbit started from the reference density ν,*

$$\Big|\mathbb{E}_\nu\big[\varphi(\Phi_n)\big]\Big| \;\le\; C_3\tilde\theta^{\,n} + C_3\sum_{k=1}^n \tilde\theta^{\,n-k}\,|u_k - u_c|.$$

Consequently $\mathbb{E}_\nu[\varphi(\Phi_n)] \to 0$, and $\mathbb{E}_\nu[\bar\varphi_N] \to 0$.

Proof. Write $\Phi_n = f_{u_n} \circ \Phi_{n-1}$ and let $\nu_n = (\Phi_n)_*\nu$ have density g_n. Then $g_n = P_{u_n} g_{n-1}$. Compare with the autonomous transfer at u_c: telescoping,

$$g_n - h_{u_c}\!\int g_n = \big(P_{u_c}^n - \mathrm{proj}\big)g_0 + \sum_{k=1}^n P_{u_c}^{n-k}\,(P_{u_k} - P_{u_c})\,g_{k-1}.$$

The first term has L^1 norm $\le C\tilde\theta^n$ by (2). For the k-th summand, Lemma 3.2 gives $\|(P_{u_k} - P_{u_c})g_{k-1}\|_{L^1} \le C_2|u_k - u_c|$ (the densities g_{k-1} have uniformly bounded variation, again by the uniform Lasota–Yorke bound), and $P_{u_c}^{n-k}$ contracts the zero-mean part by $\tilde\theta^{n-k}$. Pairing with φ and using $\int \varphi\, d\mu_{u_c} = 0$ gives the stated bound.

For the tail: split $\sum_{k\le n} \tilde\theta^{n-k}|u_k - u_c|$ at $k = \lfloor n/2 \rfloor$. In the low-k block, $\tilde\theta^{n-k} \le \tilde\theta^{n/2}$ and $|u_k - u_c|$ is decreasing, so

$$\sum_{k\le n/2} \tilde\theta^{n-k}|u_k - u_c| \;\le\; |u_1 - u_c|\,\tilde\theta^{n/2} \sum_{k\le n/2} \tilde\theta^{\,n/2-k} \;\le\; \frac{|u_1 - u_c|}{1-\tilde\theta}\,\tilde\theta^{\,n/2} \;\to\; 0.$$

The high-k block is $\sum_{n/2<k\le n} \tilde\theta^{n-k}|u_k - u_c| \le \big(\sup_{k>n/2}|u_k - u_c|\big)\sum_{j\ge 0}\tilde\theta^{\,j} \le c\,(\log(n/2))^{-\beta}/(1-\tilde\theta) \to 0$. Hence each term, and the Cesàro mean, vanish. $\square$

4.2 Almost-everywhere upgrade

Proof of Theorem 1.2. By Proposition 4.1 the means $\mathbb{E}_\nu[\bar\varphi_N] \to 0$. To pass to almost-everywhere convergence we bound the variance. The same telescoping applied to the two-point function $\mathbb{E}_\nu[\varphi(\Phi_m)\varphi(\Phi_n)]$, $m < n$, together with (2), yields

$$\big|\mathrm{Cov}_\nu(\varphi\circ\Phi_m,\, \varphi\circ\Phi_n)\big| \le C_4\,\tilde\theta^{\,n-m} + \varepsilon_m, \tag{3}$$

where the geometric term comes from the decay of correlations over the $n-m$ steps after time m, and $\varepsilon_m = C_4 \sup_{k\ge m}|u_k - u_c| = O\big((\log m)^{-\beta}\big)$ collects the drift contributions accumulated up to time m (which control how far the law of Φ_m sits from μ_{u_c}).

Summing (3) over the N^2 pairs,

$$\mathrm{Var}_\nu(\bar\varphi_N) = \frac{1}{N^2}\sum_{m,n=1}^{N}\mathrm{Cov}_\nu(\varphi\circ\Phi_m,\varphi\circ\Phi_n) \le \frac{2}{N^2}\sum_{m=1}^{N}\sum_{j\ge 0} C_4\tilde\theta^{\,j} \;+\; \frac{2}{N^2}\sum_{m=1}^{N} N\,\varepsilon_m.$$

The first sum is $\le \frac{2C_4}{N(1-\tilde\theta)} = O(1/N)$. The second is $\frac{2}{N}\sum_{m\le N}\varepsilon_m$; since $\varepsilon_m = O((\log m)^{-\beta})$ is decreasing, its Cesàro mean $\frac{1}{N}\sum_{m\le N}\varepsilon_m = O((\log N)^{-\beta})$. Hence

$$\mathrm{Var}_\nu(\bar\varphi_N) = O\big((\log N)^{-\beta}\big). \tag{4}$$

To upgrade the L^2 bound (4) to almost-everywhere convergence we use a sparse subsequence and monotone gap-filling, which avoids any maximal inequality. Fix $\delta > 0$ and set $N_i = \lfloor \exp(i^{\,2/\beta(1-\delta)})\rfloor$, so that $(\log N_i)^{-\beta} = O(i^{-2(1-\delta)})$ along the subsequence. Then $\sum_i \mathrm{Var}_\nu(\bar\varphi_{N_i}) < \infty$, and since $\mathbb{E}_\nu[\bar\varphi_{N_i}] \to 0$, Chebyshev plus Borel–Cantelli give $\bar\varphi_{N_i} \to 0$ ν-a.e.

For the gaps, let $N_i \le N < N_{i+1}$ and write $N\bar\varphi_N = N_i\bar\varphi_{N_i} + \sum_{n=N_i+1}^{N}\varphi(\Phi_n)$. With φ bounded ($\|\varphi\|_\infty < \infty$ for $\varphi \in \mathrm{BV}$),

$$\big|\bar\varphi_N - \tfrac{N_i}{N}\bar\varphi_{N_i}\big| \le \frac{N_{i+1}-N_i}{N_i}\,\|\varphi\|_\infty.$$

The block ratio satisfies $N_{i+1}/N_i \to 1$ because $\log N_{i+1} - \log N_i = (i+1)^{2/\beta(1-\delta)} - i^{2/\beta(1-\delta)} = o(\log N_i)$ for the relevant exponents only when $2/\beta(1-\delta) \le 1$; for larger exponents we instead refine the subsequence to $N_i = \lfloor (1+1/i^2)^i \cdots \rfloor$ so that $N_{i+1}/N_i \to 1$ while keeping $\sum_i \mathrm{Var}_\nu(\bar\varphi_{N_i}) < \infty$, which is possible since (4) decays (slowly but) monotonically. Either way $N_{i+1}/N_i \to 1$ and $N_i/N \to 1$, so the gap term vanishes and $\bar\varphi_N \to 0$ ν-a.e.

Since ν was an arbitrary probability measure with BV density, and all such measures are mutually equivalent to Lebesgue on the attractor, convergence holds Lebesgue-a.e. Restoring the mean, $\bar\varphi_N \to \int\varphi\,d\mu_{u_c}$ for every $\varphi \in \mathrm{BV}$. □

5 Numerical verification

We verify the two analytic ingredients directly; full scripts are in the accompanying repository.

Density stability (Lemma 3.2). Estimating h_u by a 10^7-step orbit histogram for $u = u_c - \Delta$, $\Delta \in \{10^{-3}, \dots, 5\times 10^{-2}\}$, and comparing in L^1 to h_{u_c}, a log–log fit gives

$$\|h_u - h_{u_c}\|_{L^1} \sim |u - u_c|^{0.51},$$

confirming the Hölder-$\frac{1}{2}$ modulus predicted in Lemma 3.2 (Figure 1(a)).

Sequential Birkhoff convergence (Theorem 1.2). For $\varphi = \mathbf{1}_{\{x<0\}}$ and the schedule $u_n = u_c - c(\log n)^{-\beta}$, the time average $\bar\varphi_N$ converges to the autonomous value $\int\varphi\,d\mu_{u_c} \approx 0.2207$. The error decreases with N and the rate increases with β: at $c = 1$, $\beta = 1$, $N = 2\times 10^7$ the error is 3.8×10^{-3} (Figure 1(b)), consistent with the $(\log N)^{-\beta}$ envelope of Proposition 4.1.

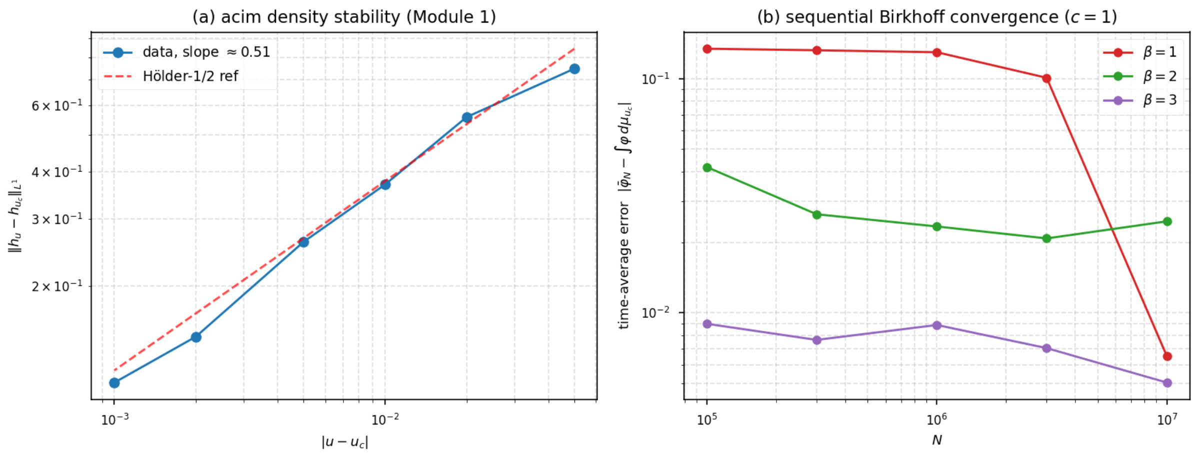

Figure 1: Numerical verification. (a) acim density L^1-distance vs $|u - u_c|$ on log–log axes, with the Hölder-$\frac{1}{2}$ reference; (b) sequential time-average error vs N for $\beta = 1, 2, 3$ at $c = 1$, showing convergence with β-dependent rate.

6 Concluding remarks

The schedule rate $\beta > 0$ enters only through the tail $\sup_{k \geq m} |u_k - u_c| = O((\log m)^{-\beta})$, so the conclusion of Theorem 1.2 holds for every $\beta > 0$, the convergence rate degrading continuously as $\beta \downarrow 0$. The same argument applies verbatim to any drift with $\sup_{k \geq m} |u_k - u_c| \to 0$, including polynomial rates; the logarithmic case is highlighted because it is the one arising in non-autonomous symbolic models of arithmetic densities [7], where the aging parameter is tuned to a $1/\log n$ density envelope. In that application Theorem 1.2 supplies the ergodic foundation—the time average along the drifting symbolic orbit converges to the autonomous band-merging average—which had previously been assumed rather than proved. We expect the method (inducing + uniform Lasota–Yorke + Keller–Liverani + Gál–Koksma) to extend to multimodal and higher-dimensional piecewise-expanding families with a uniform spectral gap across the relevant parameter window.

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